

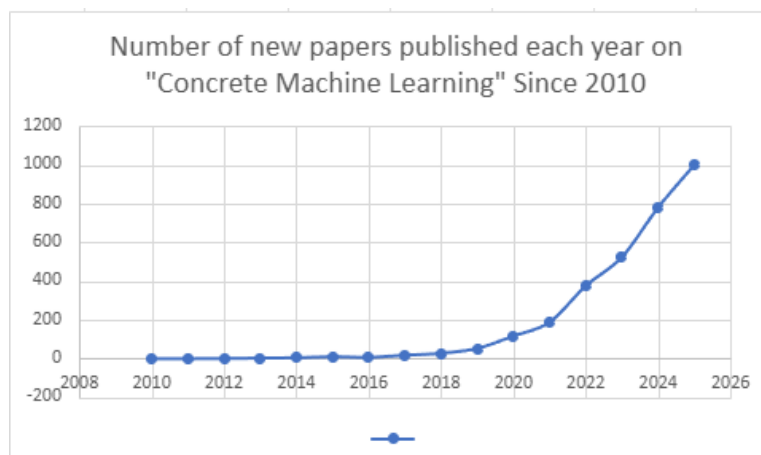
Investigation into Machine learning in concrete

Intro

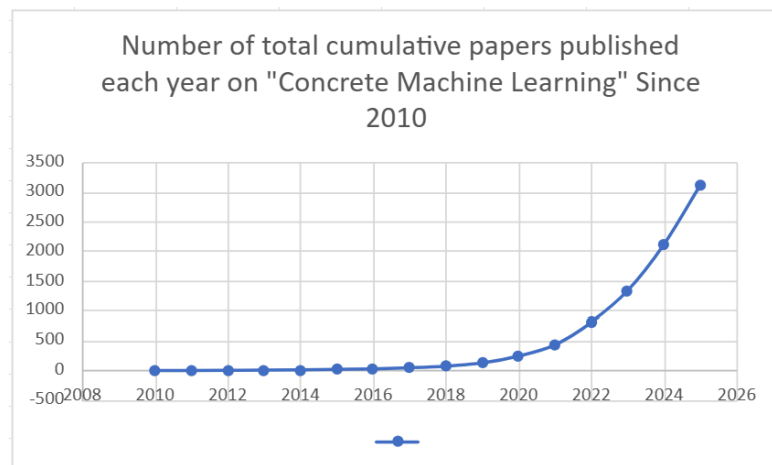
Machine learning in concrete is an interesting and rapidly developing area of research. It offers new ways to predict, analyze, and optimize the properties and performance of concrete, helping improve efficiency and accuracy in construction. As advances in artificial intelligence continue, research is increasing and changing and hopefully this report offers an overview for some of these factors.

Trends in Research quantity

Since 2010 there has been a vast increase in the number of academic papers published on the topics and relationships between concrete and Machine learning. As shown in the following graph.



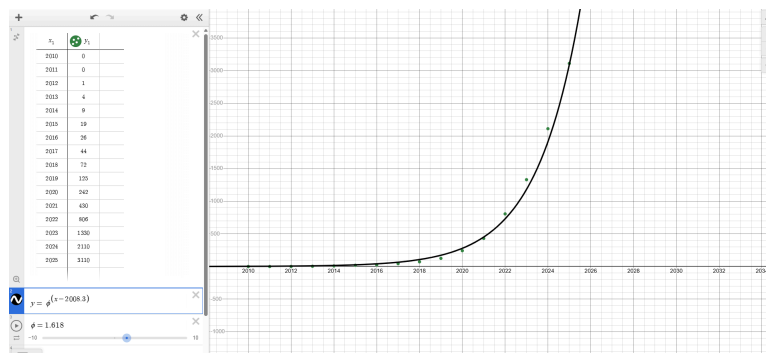
And if we plot this cumulatively we get:



which lines up well with the exponential model (only adjusted by a horizontal translation):

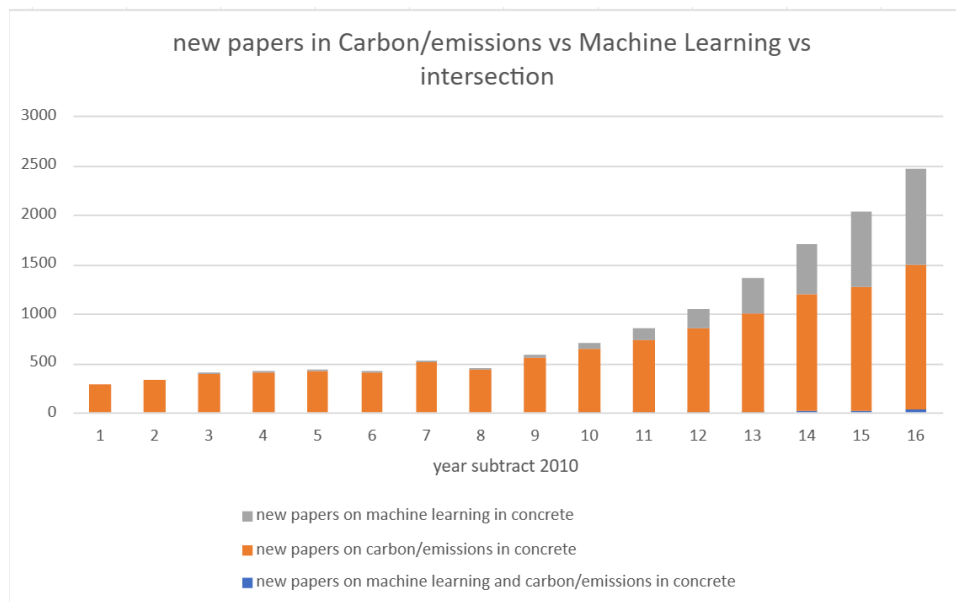
$$y = \phi^x$$

Proving how beautiful this topic and industry is

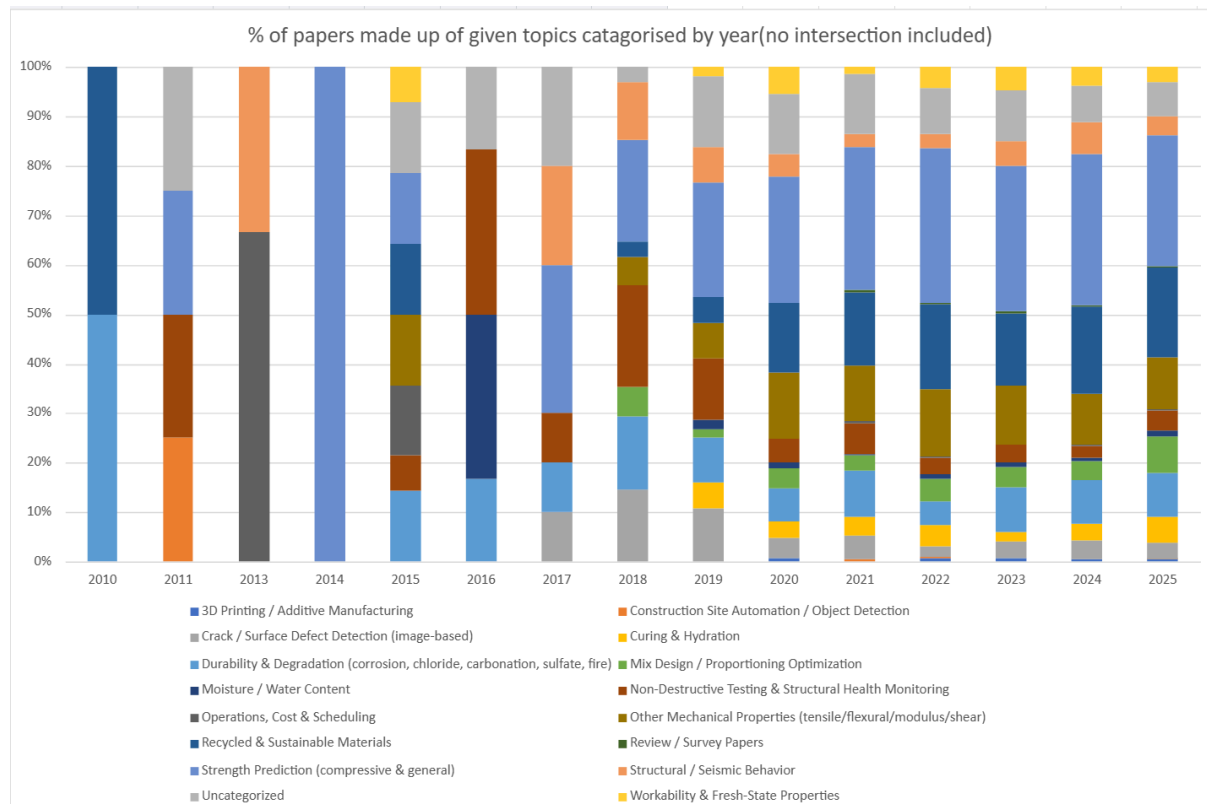


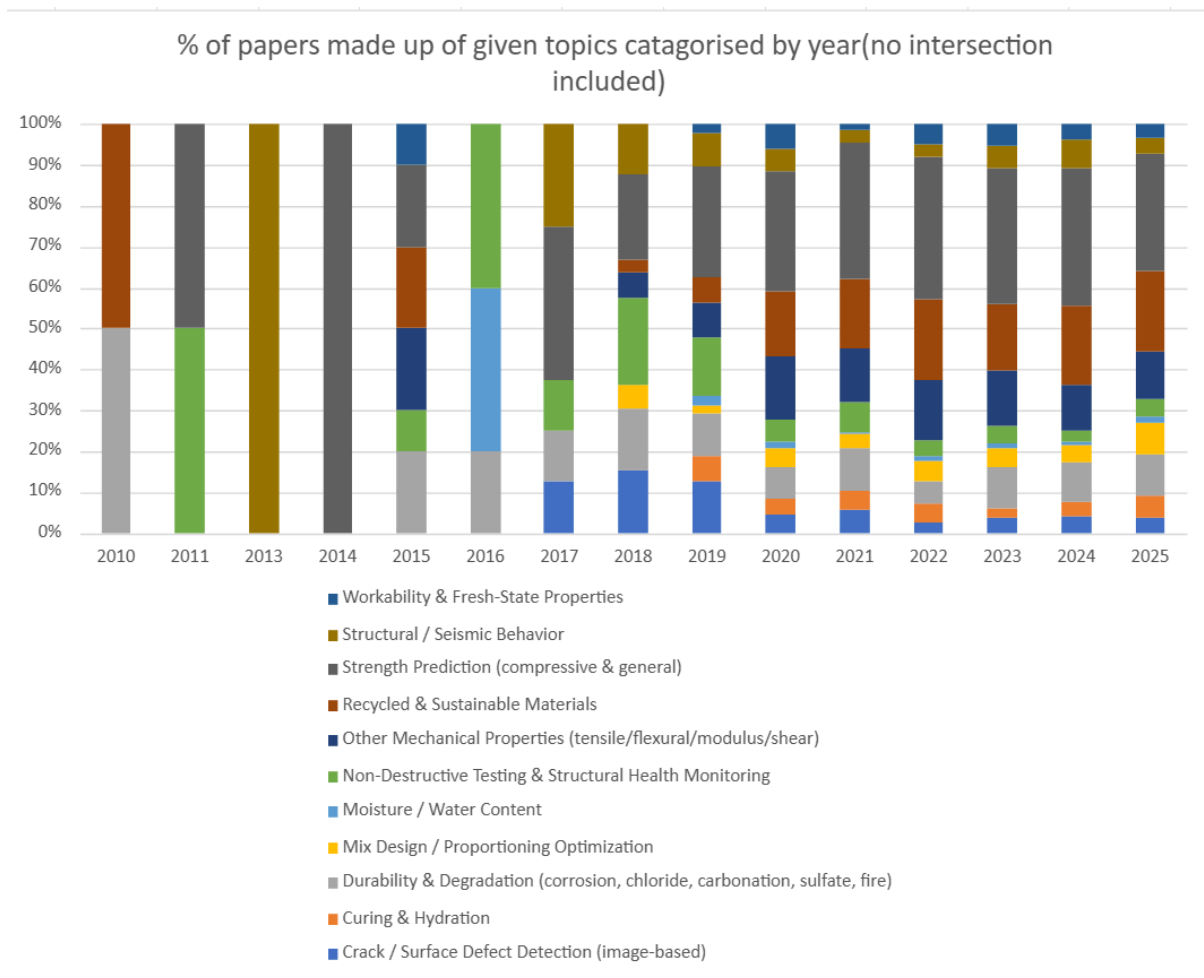
Additionally St Paul's is the building that fits to the golden ratio the most (coincidence?) <https://www.goldennumber.net/roofing-megastore-uk-worlds-most-beautiful-buildings-by-science-and-golden-ratio/>.

We can also look at the relationship between new papers in machine learning vs sustainability in the concrete industry including the small (but growing!) intersection.



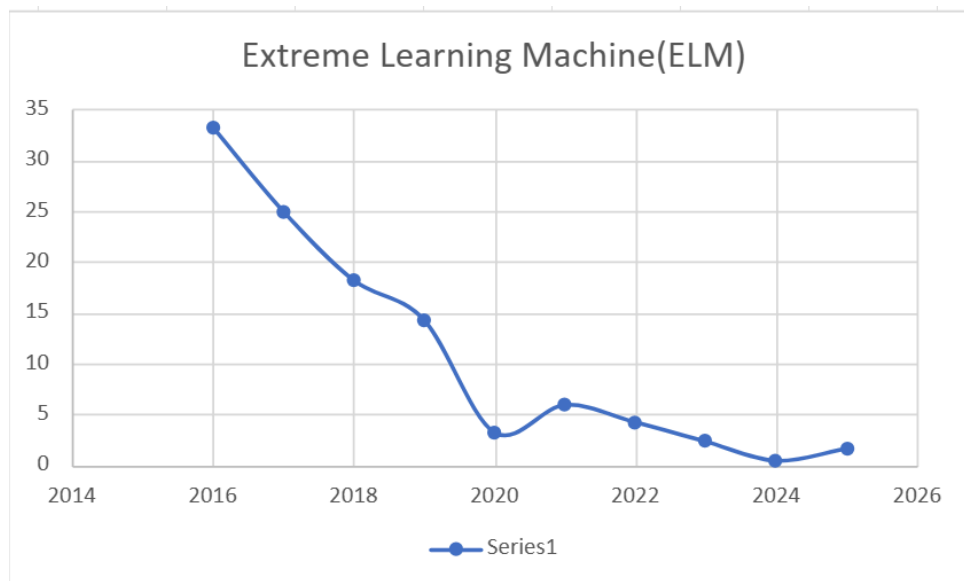
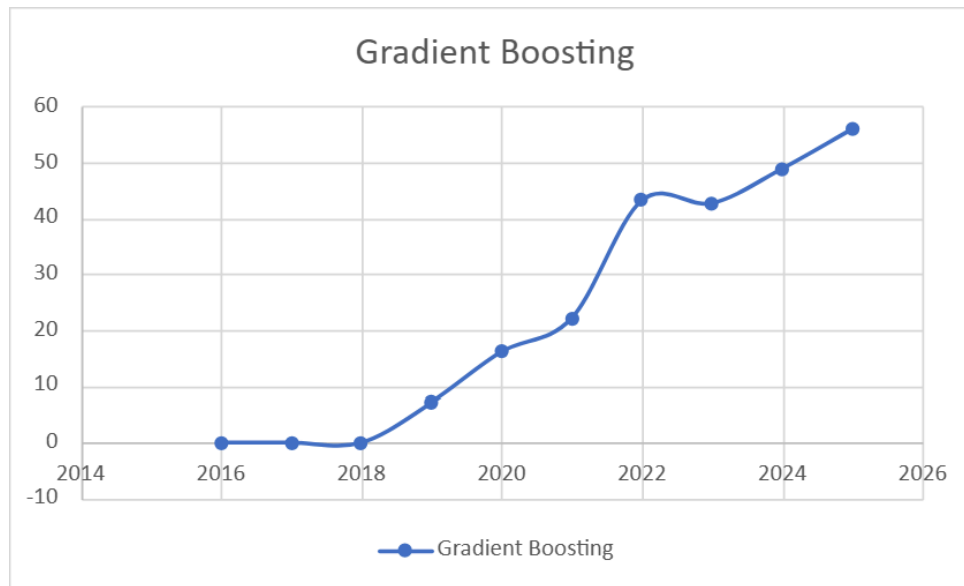
Additionally the percentage of a given topic out of the total number of topics can be found within “concrete machine learning” papers. However the earlier years are not very useful due to the very small sample size. This is the categorisation that Claude recommended followed by a simplified/reduced version which omits the smaller and less relevant categories.

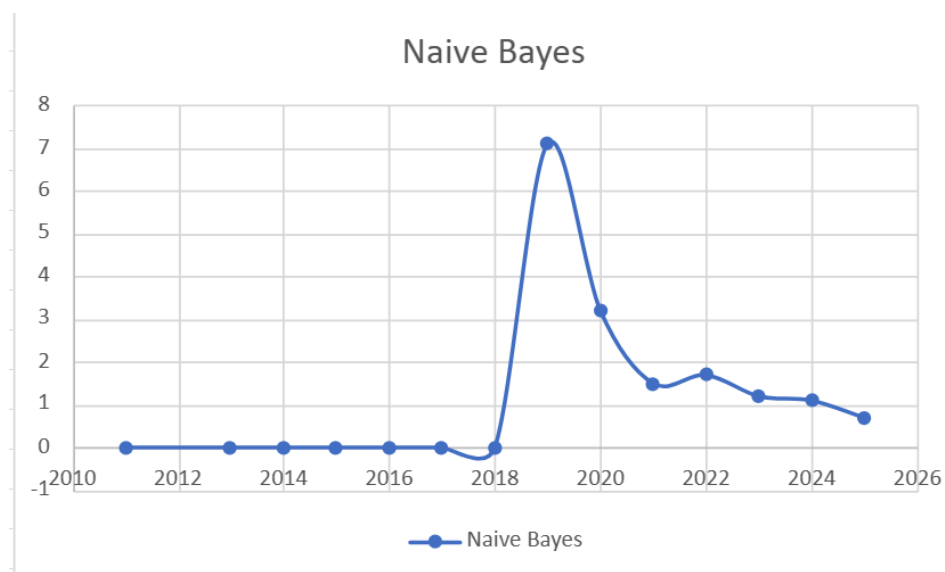
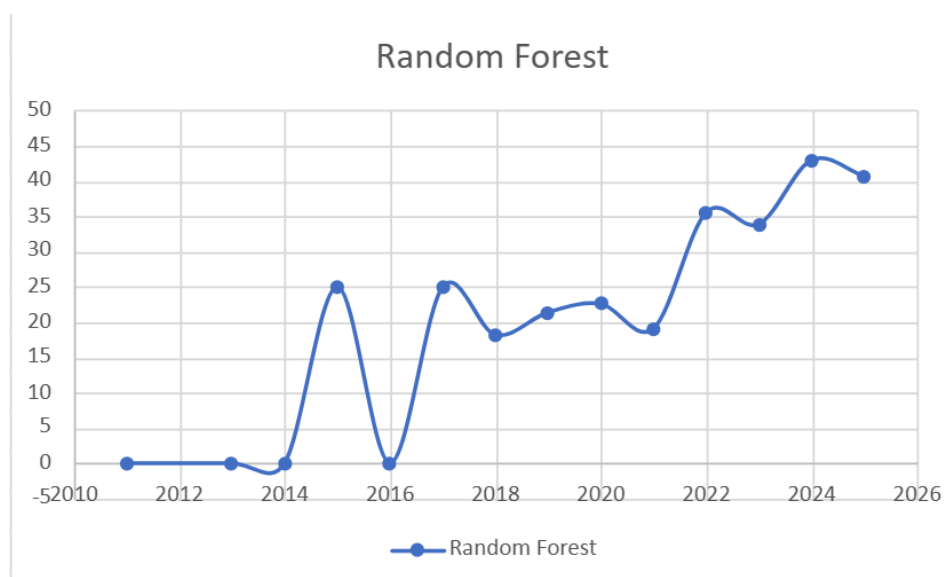
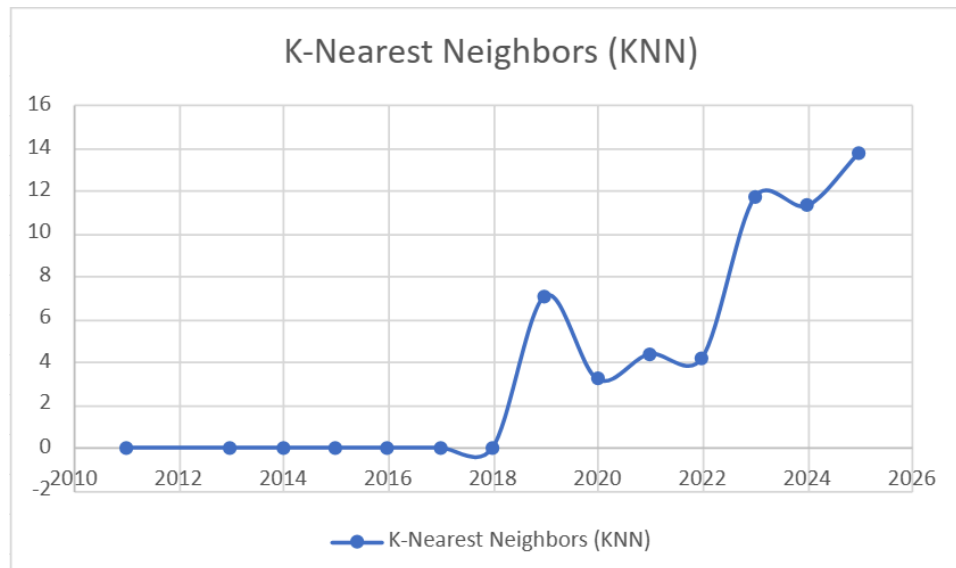


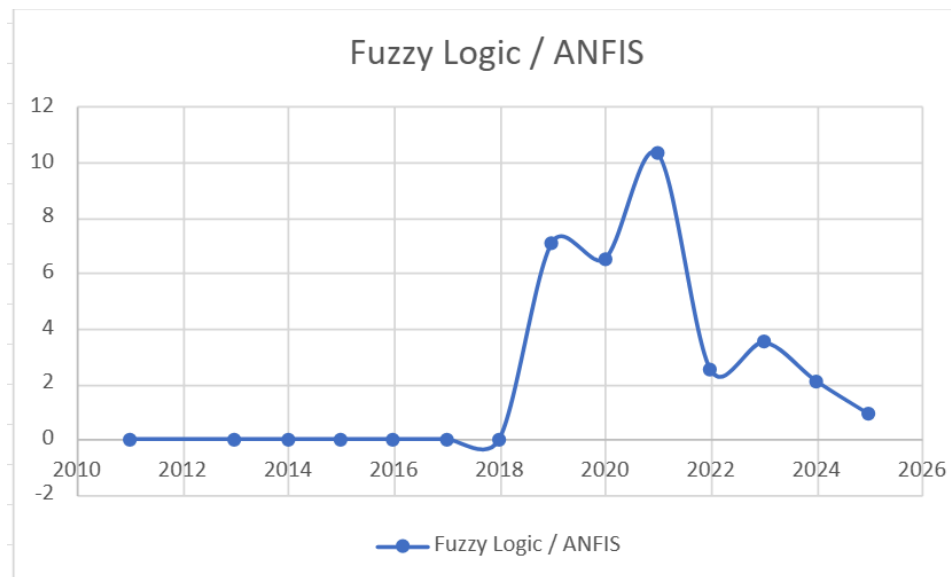


Trends in Machine Learning methods

I used a python scraper for "machine learning concrete" in openalex (an academic paper library) and matched keywords in the abstracts to find different ML methods used. I then plotted the % of each individual method that occur compared to the total number of papers with abstracts containing methods for each year. The most notable trends are shown below however some start at 2016 as before then there are very few papers with method mentioning abstracts(literally single digits) and a single paper with the method creates a huge anomaly ruining the trend (you can see this in 2015 for random forest). Big thanks to Kiran for checking over this (and also identifying the trend of fuzzy logic as suspicious).

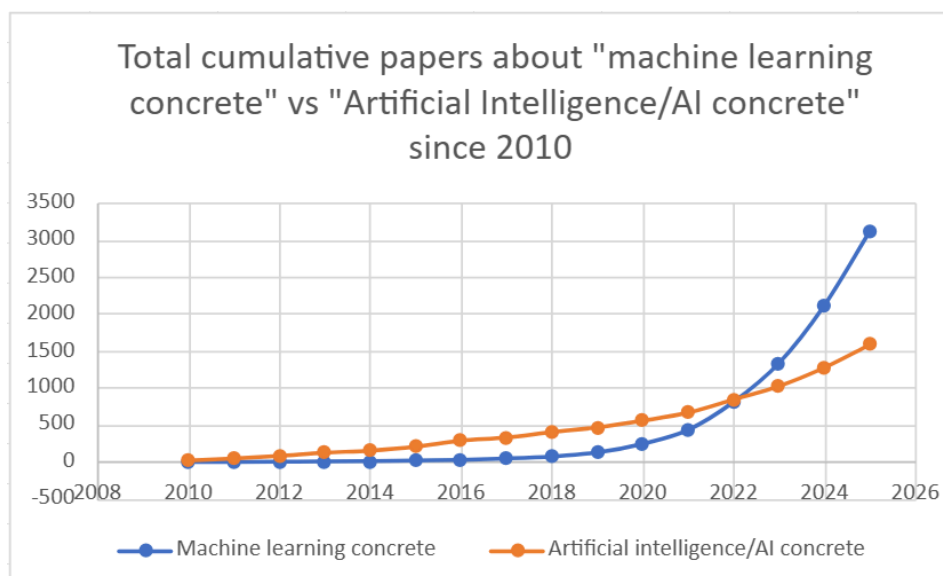


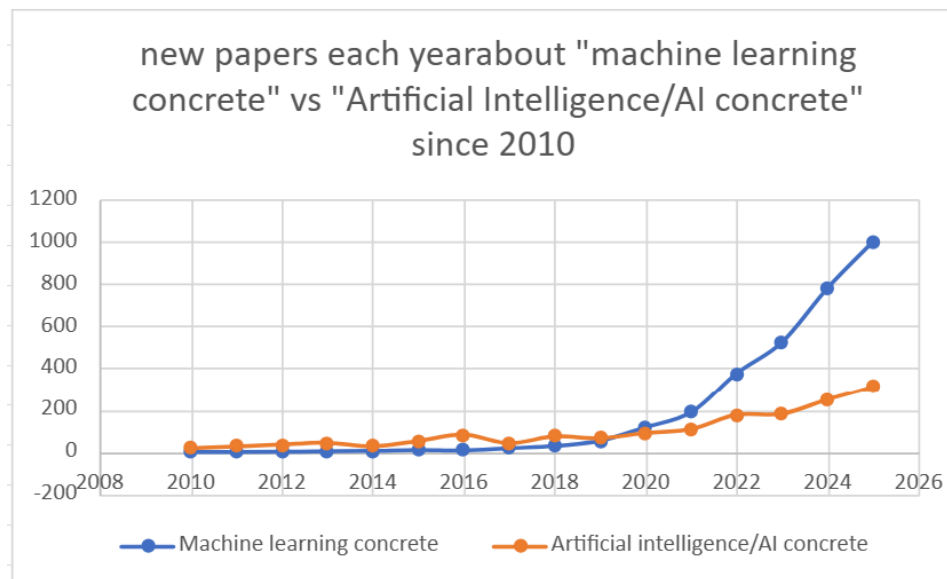




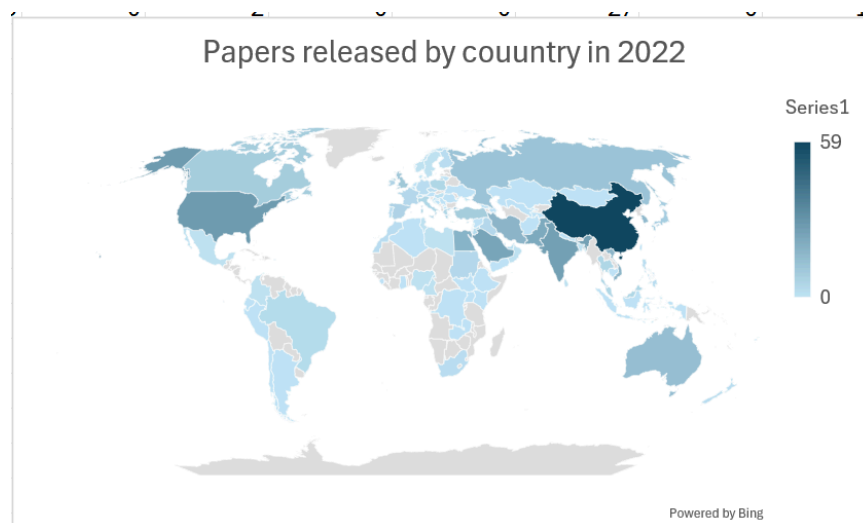
Miscellaneous

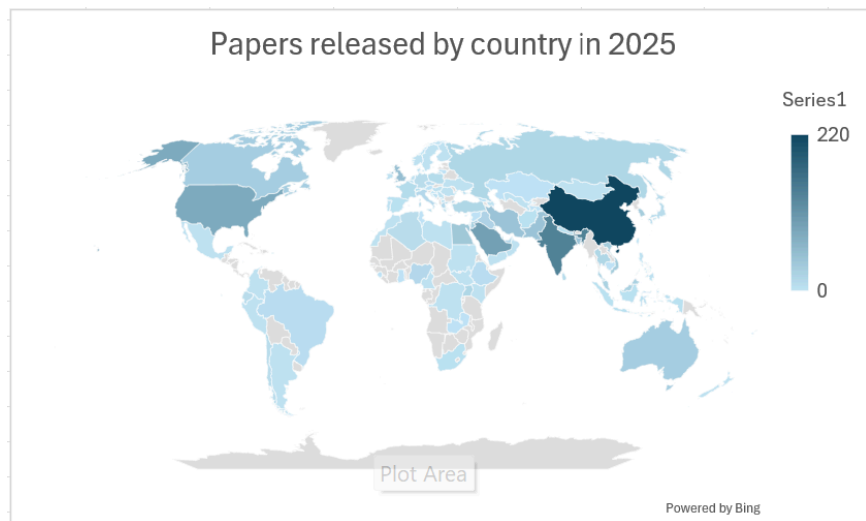
I also thought It would be interesting to compare AI vs Machine learning in the titles of papers. As shown below. I originally thought that AI would be a more modern phrase in comparison to machine learning however it seems to be the reverse. Maybe this is due to "AI" being associated with LLM's or maybe something else but its interesting to see a clear trend for something I imagine to be subconsciously influenced.





I also thought It would be interesting to show the output of papers in this field based on geographical location so I used 2 years with large numbers of papers (so hopefully representative) to display this. And as you can see China is the power house of academic research on the topic of ML in concrete.





Remarks

The majority of what I produced over the first 2 days wasn't used in this report since I shifted my focus so that I could compare the whole space.

Additionally in hindsight I've realised that included in some of these statistics are citations which add the count despite not being papers however the general trend should remain the same regardless just with a slightly smaller magnitude.

Many thanks to everyone at converge, I have really enjoyed this work experience!

Thank you to Gideon for helping me through this project, Alex for the safeguarding talk, Maria for being amazing, Vivek for the construction talk, Kiran for calling to help, Guillaume for giving me ideas and Max for teaching me how to tim tam slam.